Causal Analysis Report

BPI_Challenge_2013_open_problems.xes.gz-im

Decision points:	4702
Choice sets:	2
Transitions fitted:	5
Significant (p<0.05):	0
Generated:	2026-06-21 12:58

Step 1 — Association test (logistic regression + permutation)
H_0: choice ⊥ util | choice_set.   reject_H0=True → util predicts routing.

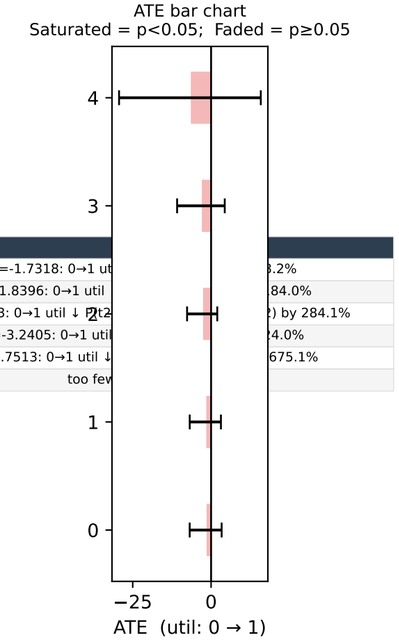
(no data)

Step 2 — Double ML (partially linear regression, HC3 robust SE)
θ = linear effect of 1-unit ↑ util on P(chose k), after partialling out X.
Equals ATE when the effect is homogeneous. Otherwise: variance-weighted average of CATEs.

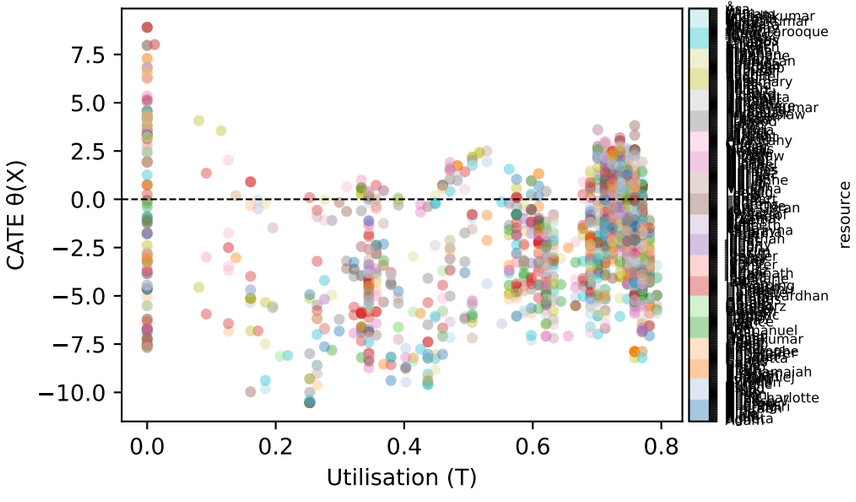
choice_set	choice_set_labels	n	choices_seen	transition	transition_label	alt_utils	theta	se	ci_95	p_value	reject_H0	note
(0, 1, 2, 3, 4)	('Wait_t0', 'Assigned_t1', 'Awaiting Assignment_t2', 'Closed_t3', 'In Progress_t4')	2351	0, 1, 2, 3, 4	0	Wait_t0	Assigned_t1, Awaiting Assignment_t2, Closed_t3, In Progress_t4	0.626536	0.269218	[0.0989, 1.1542]	0.02	True	θ=+0.6265: 1-unit ↑ util ↑ P(t0/Wait_t0) by 62.7%
(0, 1, 2, 3, 4)	('Wait_t0', 'Assigned_t1', 'Awaiting Assignment_t2', 'Closed_t3', 'In Progress_t4')	2351	0, 1, 2, 3, 4	1	Assigned_t1	Awaiting Assignment_t2, Closed_t3, In Progress_t4, Wait_t0	0.381463	0.371347	[-0.3464, 1.1093]	0.3043	False	θ=+0.3815: 1-unit ↑ util ↑ P(t1/Assigned_t1) by 38.1%
(0, 1, 2, 3, 4)	('Wait_t0', 'Assigned_t1', 'Awaiting Assignment_t2', 'Closed_t3', 'In Progress_t4')	2351	0, 1, 2, 3, 4	2	Awaiting Assignment_t2	Assigned_t1, Closed_t3, In Progress_t4, Wait_t0	-0.496871	0.277255	[-1.0403, 0.0465]	0.0731	False	θ=-0.4969: 1-unit ↑ util ↓ P(t2/Awaiting Assignment_t2) by 49.7%
(0, 1, 2, 3, 4)	('Wait_t0', 'Assigned_t1', 'Awaiting Assignment_t2', 'Closed_t3', 'In Progress_t4')	2351	0, 1, 2, 3, 4	3	Closed_t3	Assigned_t1, Awaiting Assignment_t2, In Progress_t4, Wait_t0	1.219292	0.397992	[0.4392, 1.9994]	0.0022	True	θ=+1.2193: 1-unit ↑ util ↑ P(t3/Closed_t3) by 121.9%
(0, 1, 2, 3, 4)	('Wait_t0', 'Assigned_t1', 'Awaiting Assignment_t2', 'Closed_t3', 'In Progress_t4')	2351	0, 1, 2, 3, 4	4	In Progress_t4	Assigned_t1, Awaiting Assignment_t2, Closed_t3, Wait_t0	1.897692	0.835531	[0.2601, 3.5353]	0.0231	True	θ=+1.8977: 1-unit ↑ util ↑ P(t4/In Progress_t4) by 189.8%
(5, 6)		0		—	—	—	nan	nan	—	nan	—	too few observations

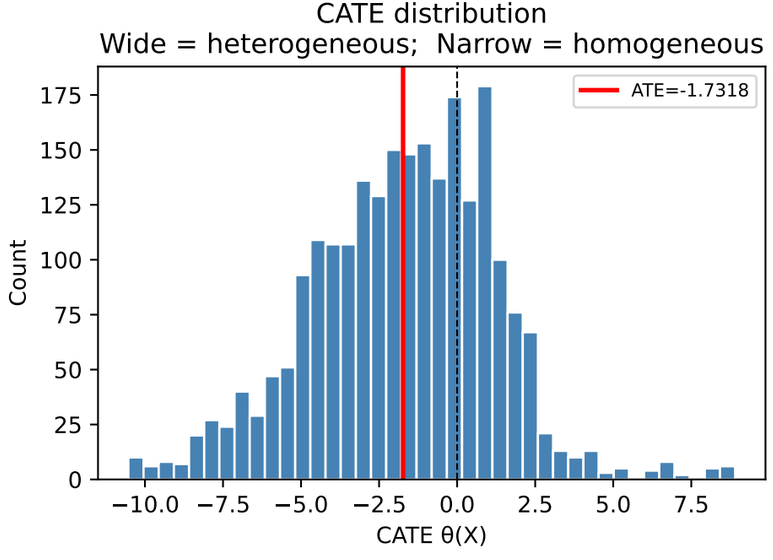
Step 3 — Causal Forest DML (ATE per transition)
ATE = mean of per-sample CATEs. Green rows = H0 rejected (p<0.05).
Less powerful for significance than Double ML, but basis for heterogeneity analysis.

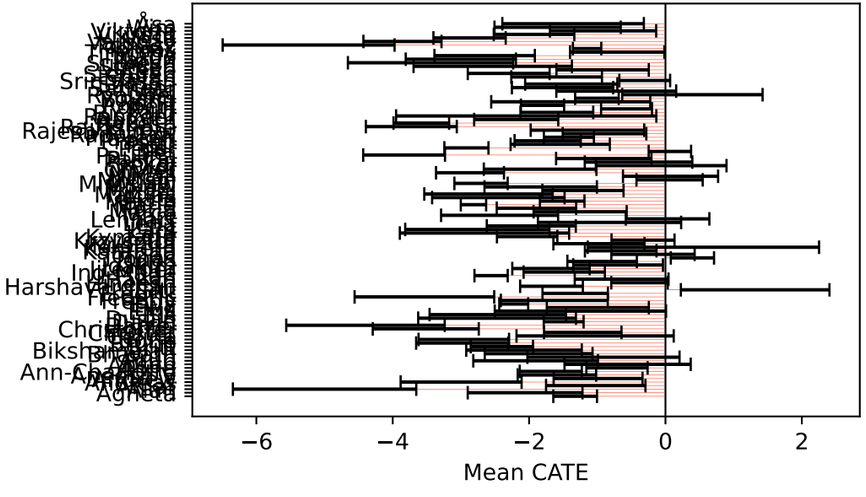
choice_set	choice_set_labels	n	choices_seen	transition	transition_label	alt_utils	ate	ci_95	p_value	reject_H0	
(0, 1, 2, 3, 4)	('Wait_t0', 'Assigned_t1', 'Awaiting Assignment_t2', 'Closed_t3', 'In Progress_t4')	2351	0, 1, 2, 3, 4	0	Wait_t0	Assigned_t1, Awaiting Assignment_t2, Closed_t3, In Progress_t4	-1.731802	[-6.8297, 3.3661]	0.5055	False	ATE=-1.7318: 0→1 ut
(0, 1, 2, 3, 4)	('Wait_t0', 'Assigned_t1', 'Awaiting Assignment_t2', 'Closed_t3', 'In Progress_t4')	2351	0, 1, 2, 3, 4	1	Assigned_t1	Awaiting Assignment_t2, Closed_t3, In Progress_t4, Wait_t0	-1.839602	[-6.8181, 3.1389]	0.4689	False	ATE=-1.8396: 0→1 util
(0, 1, 2, 3, 4)	('Wait_t0', 'Assigned_t1', 'Awaiting Assignment_t2', 'Closed_t3', 'In Progress_t4')	2351	0, 1, 2, 3, 4	2	Awaiting Assignment_t2	Assigned_t1, Closed_t3, In Progress_t4, Wait_t0	-2.841251	[-7.6565, 1.9740]	0.2475	False	ATE=-2.8413: 0→1 util ↓ P(t2
(0, 1, 2, 3, 4)	('Wait_t0', 'Assigned_t1', 'Awaiting Assignment_t2', 'Closed_t3', 'In Progress_t4')	2351	0, 1, 2, 3, 4	3	Closed_t3	Assigned_t1, Awaiting Assignment_t2, In Progress_t4, Wait_t0	-3.240466	[-10.8386, 4.3576]	0.4032	False	ATE=-3.2405: 0→1 util
(0, 1, 2, 3, 4)	('Wait_t0', 'Assigned_t1', 'Awaiting Assignment_t2', 'Closed_t3', 'In Progress_t4')	2351	0, 1, 2, 3, 4	4	In Progress_t4	Assigned_t1, Awaiting Assignment_t2, Closed_t3, Wait_t0	-6.751276	[-29.3554, 15.8529]	0.5583	False	ATE=-6.7513: 0→1 util ↓
(5, 6)		0		—	—	—	nan	—	nan	—	too few

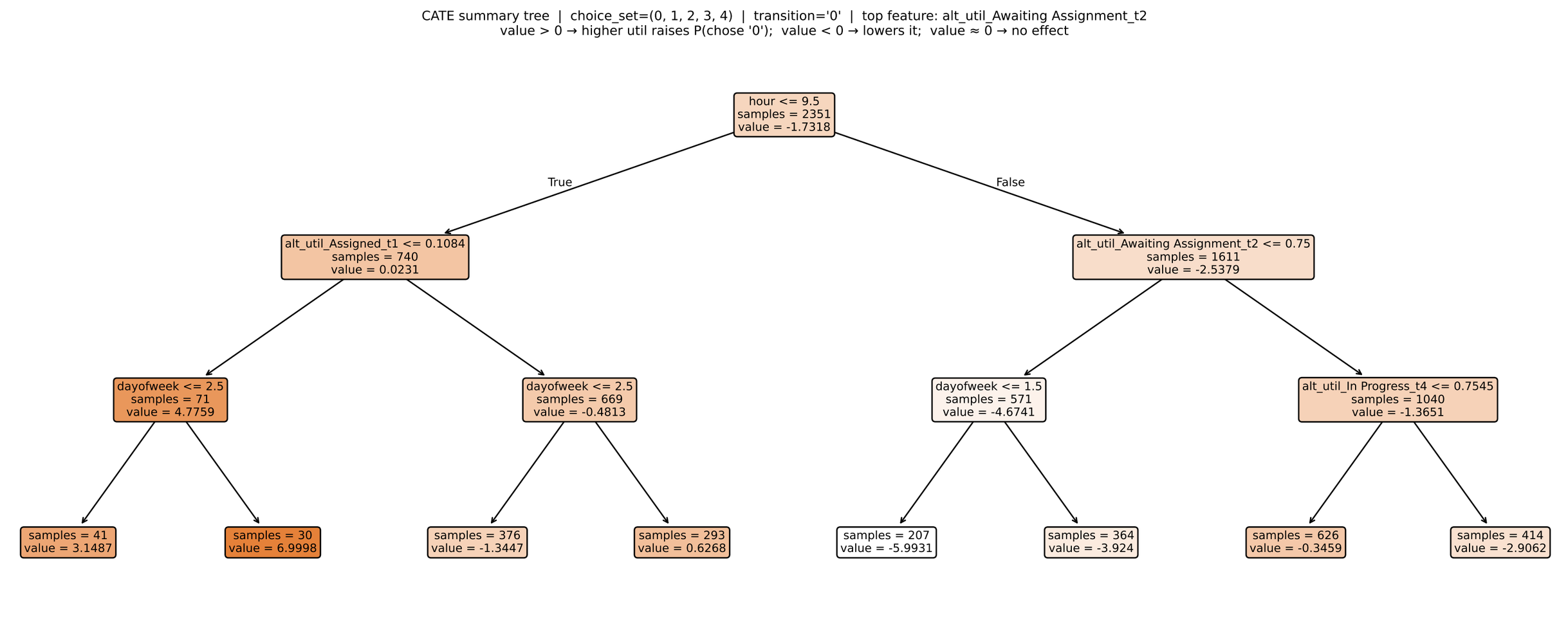


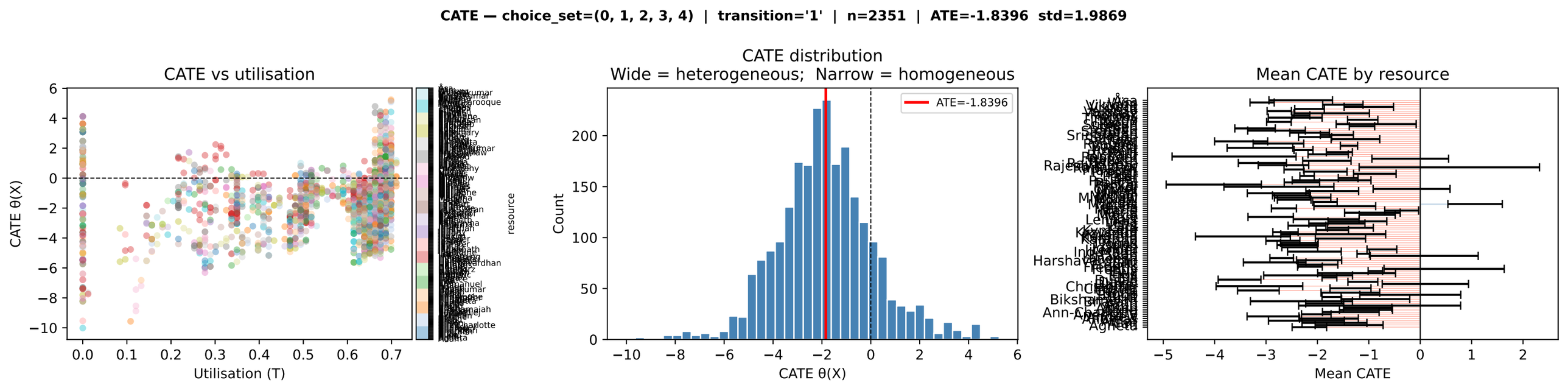
CATE — choice_set=(0, 1, 2, 3, 4) | transition='0' | n=2351 | ATE=-1.7318 std=2.9456

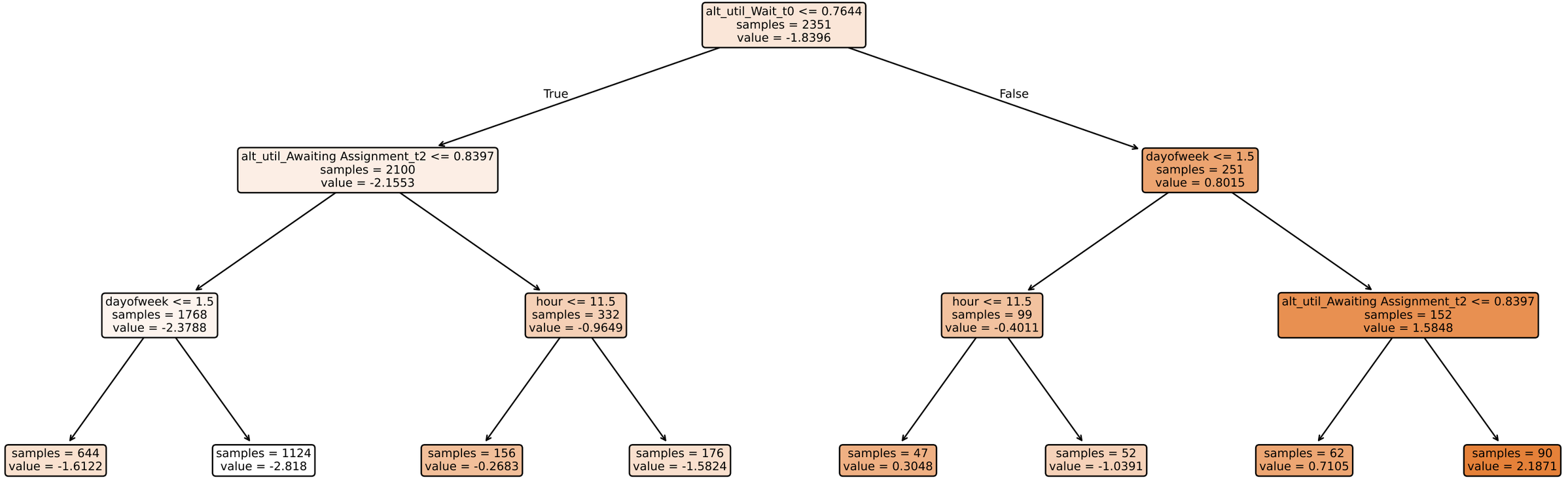




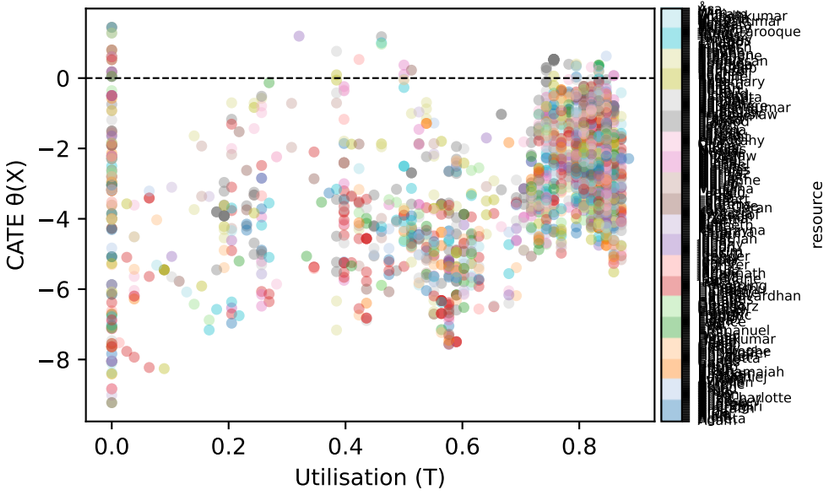


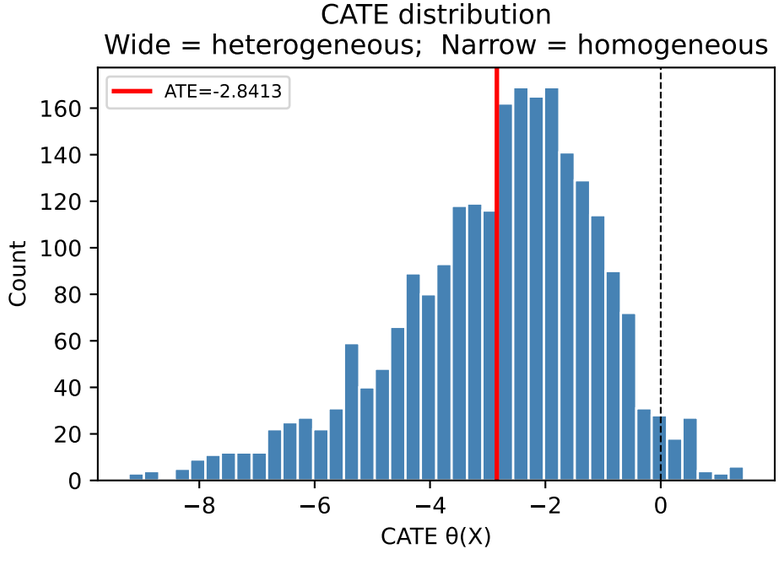

CATE summary tree | choice_set=(0, 1, 2, 3, 4) | transition='0' | top feature: alt_util_Awaiting Assignment_t2
value > 0 → higher util raises P(chose '0'); value < 0 → lowers it; value ≈ 0 → no effect
hour <= 9.5
samples = 2351
value = -1.7318
True
False
alt_util_Assigned_t1 <= 0.1084
samples = 740
value = 0.0231
alt_util_Awaiting Assignment_t2 <= 0.75
samples = 1611
value = -2.5379
dayofweek <= 2.5
samples = 71
value = 4.7759
dayofweek <= 2.5
samples = 669
value = -0.4813
dayofweek <= 1.5
samples = 571
value = -4.6741
alt_util_ln Progress_t4 <= 0.7545
samples = 1040
value = -1.3651
samples = 41
value = 3.1487
samples = 30
value = 6.9998
samples = 376
value = -1.3447
samples = 293
value = 0.6268
samples = 207
value = -5.9931
samples = 364
value = -3.924
samples = 626
value = -0.3459
samples = 414
value = -2.9062

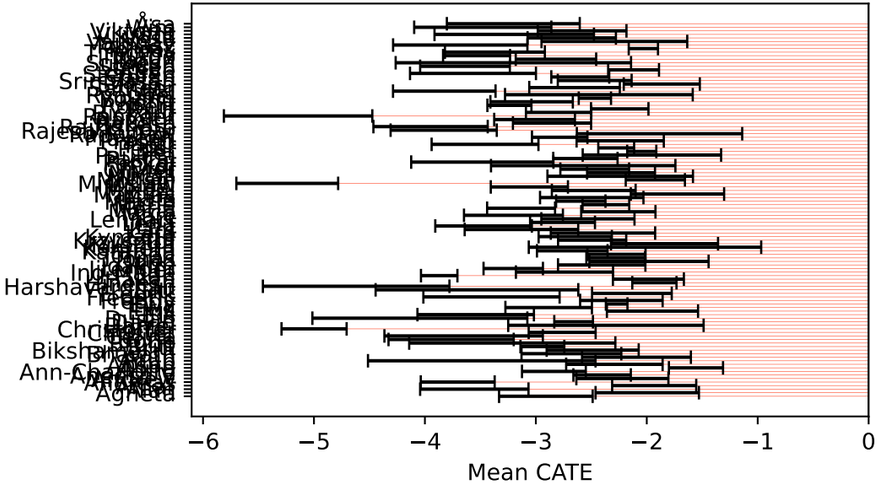

CATE — choice_set=(0, 1, 2, 3, 4) | transition='1' | n=2351 | ATE=-1.8396 std=1.9869
CATE vs utilisation
CATE θ(X)
Utilisation (T)
resource
CATE distribution
Wide = heterogeneous; Narrow = homogeneous
ATE=-1.8396
Count
CATE θ(X)
Mean CATE by resource
Mean CATE

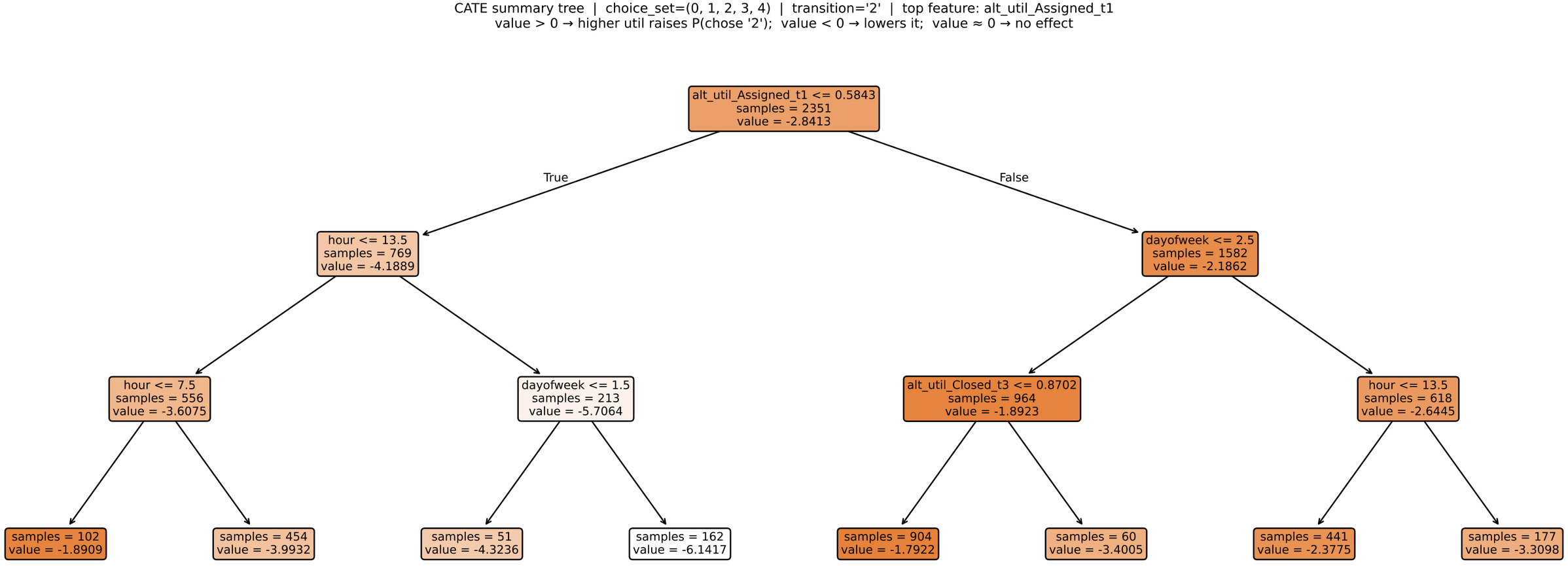

CATE summary tree | choice_set=(0, 1, 2, 3, 4) | transition='1' | top feature: alt_util_Wait_t0
value > 0 → higher util raises P(chose '1'); value < 0 → lowers it; value ≈ 0 → no effect
alt_util_Wait_t0 <= 0.7644
samples = 2351
value = -1.8396
True
False
alt_util_Awaiting Assignment_t2 <= 0.8397
samples = 2100
value = -2.1553
dayofweek <= 1.5
samples = 251
value = 0.8015
dayofweek <= 1.5
samples = 1768
value = -2.3788
hour <= 11.5
samples = 332
value = -0.9649
hour <= 11.5
samples = 99
value = -0.4011
alt_util_Awaiting Assignment_t2 <= 0.8397
samples = 152
value = 1.5848
samples = 644
value = -1.6122
samples = 1124
value = -2.818
samples = 156
value = -0.2683
samples = 176
value = -1.5824
samples = 47
value = 0.3048
samples = 52
value = -1.0391
samples = 62
value = 0.7105
samples = 90
value = 2.1871

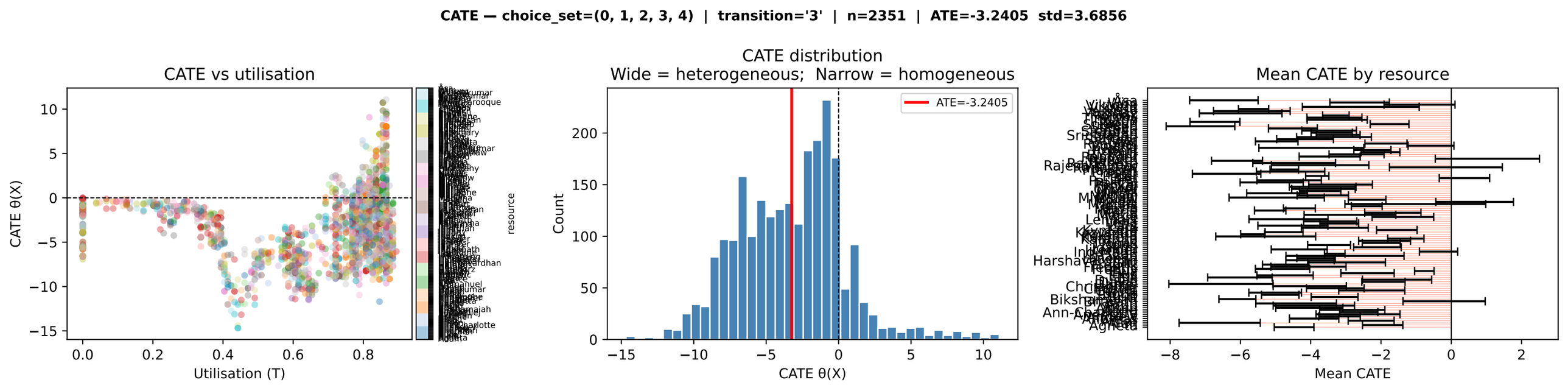
CATE — choice_set=(0, 1, 2, 3, 4) | transition='2' | n=2351 | ATE=-2.8413 std=1.7679

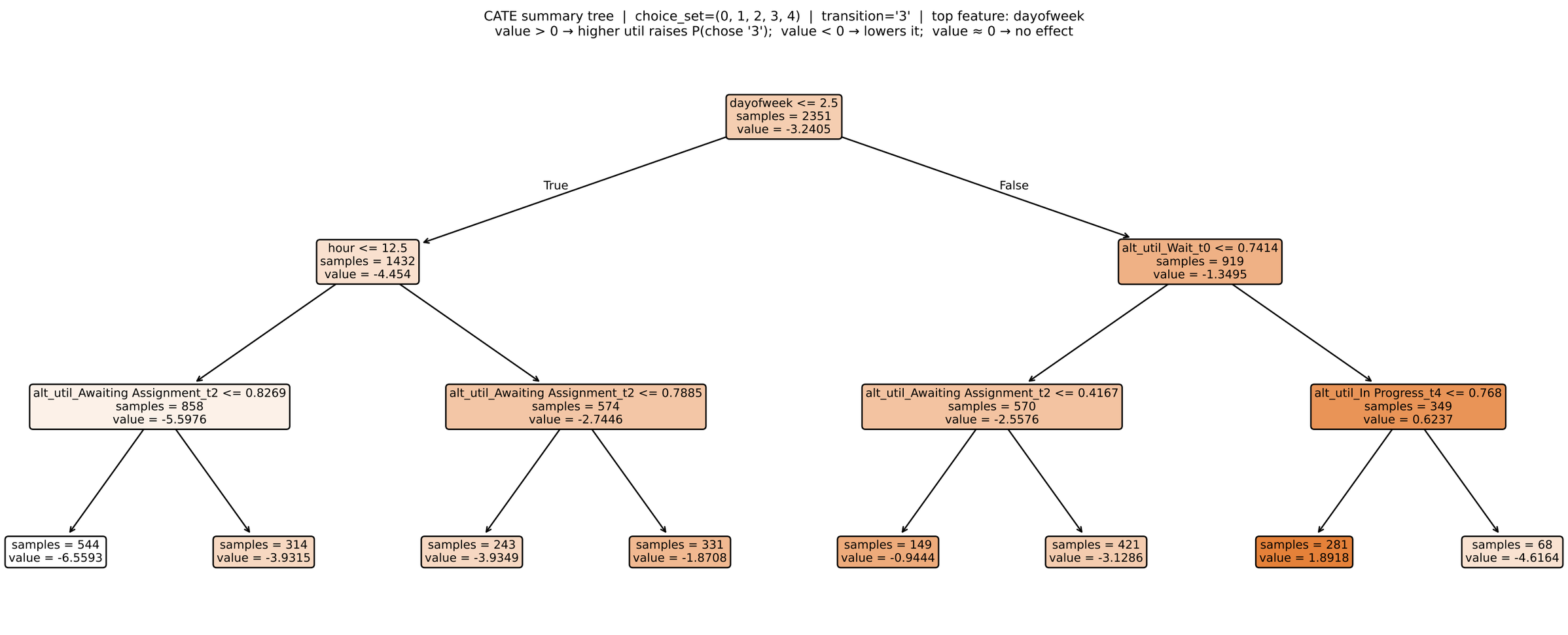


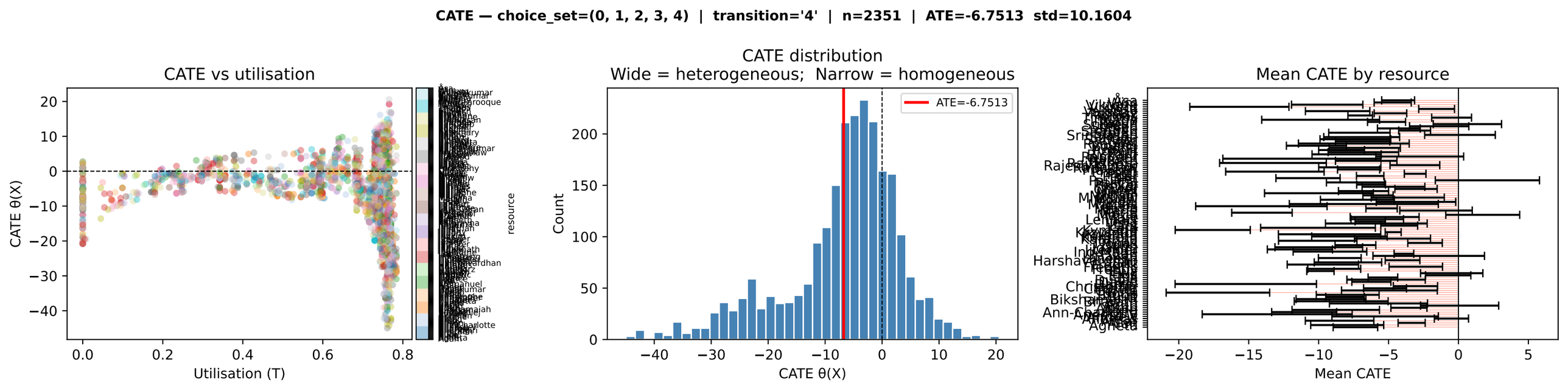


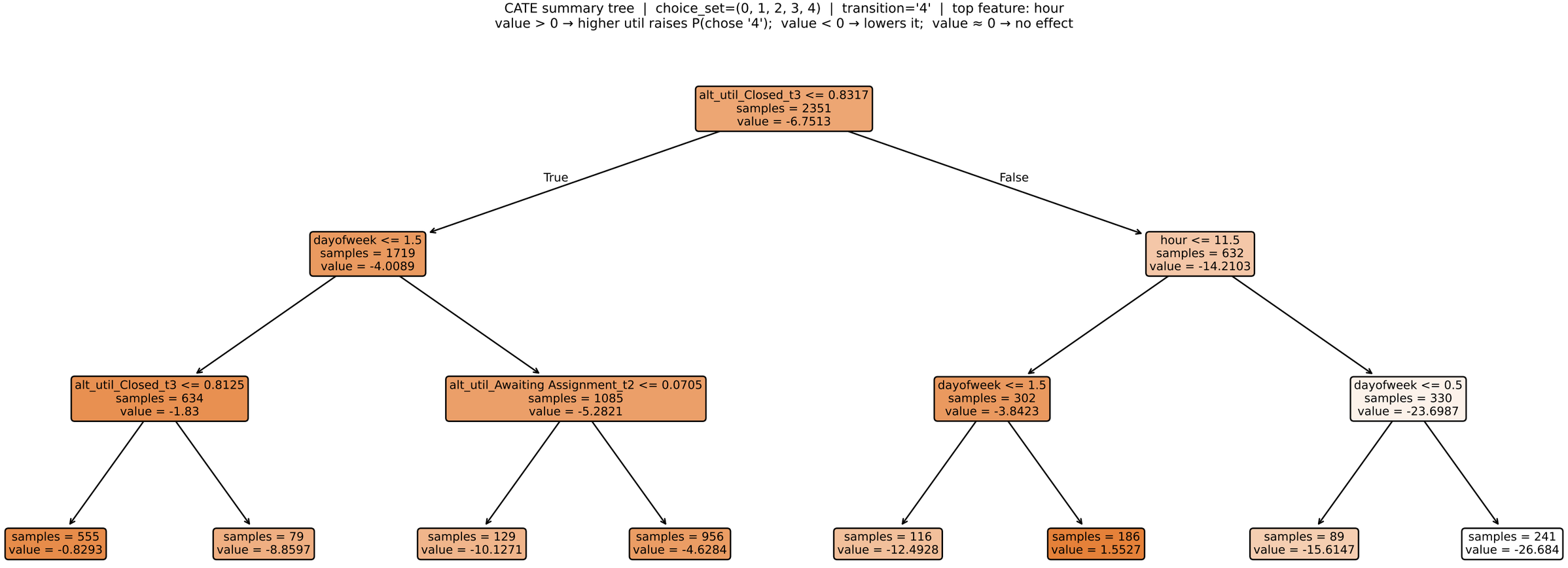



CATE summary tree | choice_set=(0, 1, 2, 3, 4) | transition='2' | top feature: alt_util_Assigned_t1
value > 0 → higher util raises P(chose '2'); value < 0 → lowers it; value ≈ 0 → no effect
alt_util_Assigned_t1 <= 0.5843
samples = 2351
value = -2.8413
True
False
hour <= 13.5
samples = 769
value = -4.1889
dayofweek <= 2.5
samples = 1582
value = -2.1862
hour <= 7.5
samples = 556
value = -3.6075
dayofweek <= 1.5
samples = 213
value = -5.7064
alt_util_Closed_t3 <= 0.8702
samples = 964
value = -1.8923
hour <= 13.5
samples = 618
value = -2.6445
samples = 102
value = -1.8909
samples = 454
value = -3.9932
samples = 51
value = -4.3236
samples = 162
value = -6.1417
samples = 904
value = -1.7922
samples = 60
value = -3.4005
samples = 441
value = -2.3775
samples = 177
value = -3.3098


CATE — choice_set=(0, 1, 2, 3, 4) | transition='3' | n=2351 | ATE=-3.2405 std=3.6856
CATE vs utilisation
CATE θ(X)
Utilisation (T)
resource
CATE distribution
Wide = heterogeneous; Narrow = homogeneous
ATE=-3.2405
Count
CATE θ(X)
Mean CATE by resource
Mean CATE


CATE summary tree | choice_set=(0, 1, 2, 3, 4) | transition='3' | top feature: dayofweek
value > 0 → higher util raises P(chose '3'); value < 0 → lowers it; value ≈ 0 → no effect
dayofweek <= 2.5
samples = 2351
value = -3.2405
True
False
hour <= 12.5
samples = 1432
value = -4.454
alt_util_Wait_t0 <= 0.7414
samples = 919
value = -1.3495
alt_util_Awaiting Assignment_t2 <= 0.8269
samples = 858
value = -5.5976
alt_util_Awaiting Assignment_t2 <= 0.7885
samples = 574
value = -2.7446
alt_util_Awaiting Assignment_t2 <= 0.4167
samples = 570
value = -2.5576
alt_util_In Progress_t4 <= 0.768
samples = 349
value = 0.6237
samples = 544
value = -6.5593
samples = 314
value = -3.9315
samples = 243
value = -3.9349
samples = 331
value = -1.8708
samples = 149
value = -0.9444
samples = 421
value = -3.1286
samples = 281
value = 1.8918
samples = 68
value = -4.6164


CATE — choice_set=(0, 1, 2, 3, 4) | transition='4' | n=2351 | ATE=-6.7513 std=10.1604
CATE vs utilisation
CATE θ(X)
Utilisation (T)
resource
CATE distribution
Wide = heterogeneous; Narrow = homogeneous
ATE=-6.7513
Count
CATE θ(X)
Mean CATE by resource
Mean CATE


CATE summary tree | choice_set=(0, 1, 2, 3, 4) | transition='4' | top feature: hour
value > 0 → higher util raises P(chose '4'); value < 0 → lowers it; value ≈ 0 → no effect
alt_util_Closed_t3 <= 0.8317
samples = 2351
value = -6.7513
True
False
dayofweek <= 1.5
samples = 1719
value = -4.0089
hour <= 11.5
samples = 632
value = -14.2103
alt_util_Closed_t3 <= 0.8125
samples = 634
value = -1.83
alt_util_Awaiting Assignment_t2 <= 0.0705
samples = 1085
value = -5.2821
dayofweek <= 1.5
samples = 302
value = -3.8423
dayofweek <= 0.5
samples = 330
value = -23.6987
samples = 555
value = -0.8293
samples = 79
value = -8.8597
samples = 129
value = -10.1271
samples = 956
value = -4.6284
samples = 116
value = -12.4928
samples = 186
value = 1.5527
samples = 89
value = -15.6147
samples = 241
value = -26.684

CATE statistics summary
cate_std = heterogeneity; pct_positive/negative = share of cases where util helps/hurts; top_feature = main moderator

choice_set	transition	n	ate	cate_std	cate_min	cate_p25	cate_p50	cate_p75	cate_max	pct_positive	pct_negative	top_feature
(0, 1, 2, 3, 4)	0	2351	-1.731802	2.945575	-10.53207	-3.691762	-1.498017	0.345508	8.904915	30.2	69.8	alt_util_Awaiting Assignment_t2
(0, 1, 2, 3, 4)	1	2351	-1.839602	1.986856	-10.018008	-3.002636	-1.903488	-0.723194	5.255873	15.7	84.3	alt_util_Wait_t0
(0, 1, 2, 3, 4)	2	2351	-2.841251	1.767862	-9.226838	-3.856905	-2.573616	-1.627192	1.443278	3.0	97.0	alt_util_Assigned_t1
(0, 1, 2, 3, 4)	3	2351	-3.240466	3.685561	-14.681047	-6.045783	-2.967651	-0.845873	11.091129	13.8	86.2	dayofweek
(0, 1, 2, 3, 4)	4	2351	-6.751276	10.160365	-44.969757	-10.586808	-4.799255	-0.461422	20.574662	22.9	77.1	hour

Key Assumptions & Interpretation Notes

▸ Backdoor criterion (all methods)

X = {hour, dayofweek, n_alternatives, res_*} must block all confounding paths T ← C → Y.

Any variable that affects both util and routing must be in X. Unobserved confounders

cannot be detected and will bias all estimates.

▸ SUTVA (all methods)

One case's routing does not affect another's treatment or outcome.

Violated if resource occupation by one case changes the util seen by a concurrent case.

▸ Positivity (all methods)

Every covariate combination X must have positive probability for all util levels T.

Diagnosed by the covariate balance (SMD) in the backdoor check.

▸ Linearity (Double ML only)

θ is the best-fit linear slope. If the effect varies with X, θ is a variance-weighted

average of CATEs — valid as a summary, but not the simple mean.

Double ML is more powerful than the forest for testing 'is there any effect?'

▸ Honest splitting (Causal Forest only)

Tree structure and leaf estimates use separate subsamples. Yields valid CIs but

halves effective sample size → forest CIs are wider than Double ML's.

A significant Double ML θ with an insignificant forest ATE is not a contradiction.

▸ What the analysis cannot establish

Causal direction (assumes util → choice, not the reverse).

External validity beyond the observed covariate distribution.

Dynamic/lagged effects across cases over time.

Full documentation: ASSUMPTIONS.md